Don't Try to Tackle the Whole Problem

Find Community Leaders

Work with Community Leaders

Move in Small Steps

How to Tackle Major Problems? Muddle Through by Taking Small Incremental Steps

We should set out to **solve only a small part of a major problem**. This way, we give ourselves the opportunity to learn from this small, initial process. It makes us better equipped and more experienced when we move on to deal with the second problem, the third problem… and the 31st problem! What we do here is to create significant and innovative changes one step at a time. Muddling through is all about "incremental innovation." It is the opposite of "radical innovation," which aims to tackle the whole problem at once in one large-scale, long-lasting and expensive project.

So, how can we tackle major problems? Follow these best practices:

01 Define Your Goal
02 Focus on the People
03 Solve the Right Problem
04 Think of Everything as a System
05 Do Small & Simple Interventions
06 Wait Until the Time is Right
07 Continue to Test
08 Don't Worry About Failure
09 Keep Your Goal in Mind
10 Adapt
11 Continue to Learn

Start Learning Today:
interaction-design.org

Creative Commons BY-SA license: You are free to edit and redistribute this template, even for commercial use, **as long as you give credit to the Interaction Design Foundation**. Also, if you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.

INTERACTION DESIGN FOUNDATION
Est. 2002

Best Practice

01 Define Your Goal

First, you define the overall goal. After building empathy via research with the people you are designing for, it's appropriate to explicitly articulate the problem you've identified and that you aim to address. This definition of the problem aims to bring clarity and focus, and is often one of the most challenging parts of the process.

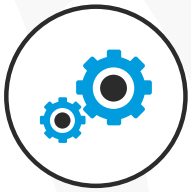
There are many methods and approaches available, including affinity diagramming and empathy mapping, but they all generally involve analyzing your data to break down complex concepts into smaller and easier to understand parts, and then filtering and creatively re-combining these data points to capture your insights about what the problems really involve as well as the beginnings of ideas for addressing them.

Ultimately, crafting a meaningful and actionable 'problem statement' is how you connect 'people centered' empathy with 'solving the right problem.' In this and any subsequent phases, you should **consider all four principles of human-centered design**:

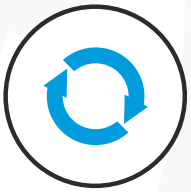
02 Focus on the People

People-Centered

Solve the Right Problem



Everything is a System



Small & Simple Interventions

People-centered: You find the community leaders and work closely with them. You study what they've already started to do. Focus on people and the context they're in in order to create things that are appropriate for these people. Work with the local people (formerly known as "users") and work within their culture. Bring in various experts from various disciplines, but let the people drive the changes. Your work with a multidisciplinary team must also include the people you're designing for, as the solution has to come from the people themselves. Otherwise, it will not work, even though external experts or designers have come in and presented the right solution. The people themselves must be part of the solution. They understand the problems they're facing, and often they have already started to create the solution to the problem at hand. As designers, we can help facilitate matters and mentor the local community, and we can bring in other resources. This is what we call "community-driven design", which is a subset of human-centered design.

Creative Commons BY-SA license: You are free to edit and redistribute this template, even for commercial use, **as long as you give credit to the Interaction Design Foundation**. Also, if you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.

03 Solve the Right Problem

Understand and solve the right problem: You do research to understand and solve the right problem, the root issues and the fundamental issues. The symptoms keep coming back if we don't solve the deep underlying causes. Figure out what's the real cause underlying the symptoms. Find and address the right problem, the underlying problem. Help the local people identify the underlying causes of their own problems.

04 Think of Everything as a System

Everything is a system: Think of everything as a system; everything is interconnected. Bring in the system approach because we know that a problem and a solution are part of a greater system. We can help the local people understand that everything is interconnected.

05 Do Small & Simple Interventions

Small and simple interventions: Do iterative work. Don't rush to a solution. Try small, simple interventions and learn from them one by one, and slowly your results will get bigger and bigger and better and better. That's the secret, but it's a hard secret when you face complex problems such as cholera epidemics, which cannot be solved overnight. Complex problems will most often take more than 10 years to solve. We must continually prototype, test and refine our proposals to make sure that our small solutions truly meet the needs of the people we're focusing on. You choose to start solving a small part of the problem while keeping in mind that it's a part of a larger problem, a larger system and just a little step towards the overall goal.

06 Wait Until the Time is Right

You **wait until the time is right** to make a very small intervention, a very small test.

07 Continue to Test

If your test is successful, you continue to solve the next little part of the major problem. You continue to do a 2nd, 3rd and—possibly—31st test and beyond, making sure that each intervention is going in the right direction.

08 Don't Worry About Failure

But what if you fail? **You can't fail**. If some of your tests are not successful, you learn from them and modify your next test-step taking your new insights into account. If your current project doesn't turn out as you planned, take it as a learning experience and modify your next initiative:

Creative Commons BY-SA license: You are free to edit and redistribute this template, even for commercial use, **as long as you give credit to the Interaction Design Foundation**. Also, if you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.

09 Keep Your Goal in Mind

Always make sure that your steps are all going in the right direction as you keep the overall final goal firmly in mind and keep adapting your small steps to reach it.

10 Adapt

Continually adapt your steps as you keep in mind that the world changes and that the problem can change along the way, as technology changes and as even the culture changes so people could be willing to do new or different things than when you took the very first steps.

11 Continue to Learn

Designers are **experienced in doing iterative work**. For designers, it's normal to try out small, simple interventions such as wireframes and paper prototypes in a more regular design context. We're used to learning from these small steps, working our way to slowly getting bigger and bigger and better and better. However, most designers are not trained to apply their human-centered design insights to these major global problems, but we can learn.

References and Where to Learn More

Norman, Donald. **The Four Fundamental Principles of Human-Centered Design and Application**. 2019.

Start Learning Today:
interaction-design.org

Creative Commons BY-SA license: You are free to edit and redistribute this template, even for commercial use, **as long as you give credit to the Interaction Design Foundation**. Also, if you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.

INTERACTION DESIGN FOUNDATION
Est. 2002

Don’t Do This: Why You Should Not Initiate a Big Radical Solution All at Once

Too Expensive

Takes Too Much Time

Disrupts Too Many People’s Lives

It’s good to be aware of the pitfalls of trying to solve major, global problems, so please check out why you should avoid initiating major radical innovative projects.

The United Nations’ 17 sustainable development problems are not new. Talented experts, competent governments and the United Nations itself have strived to solve major global problems such as hunger, lack of quality education and poor sanitation for decades. Designers can help create impactful, sustainable results which previous attempts costing billions of dollars and involving top-notch experts have not succeeded in generating.

Here’s why the major projects did not succeed. We’ve presented it all as a checklist of what you should avoid doing.

Start Learning Today:
interaction-design.org

Creative Commons BY-SA license: You are free to edit and redistribute this template, even for commercial use, **as long as you give credit to the Interaction Design Foundation**. Also, if you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.

Interaction Design Foundation
Est. 2002

Checklist of What You Should Avoid

- ☐ **01** Don Norman, who's building some of his insights on William Easterly's book The Tyranny of Experts, explains that experts generalize the problems and the solutions. They apply the same insights to the same problem (e.g., hunger). Therefore, they ignore the specific people, their culture, their capabilities and their environment.
- ☐ **02** Secondly, experts generally try to solve the major problem all at once by setting up one major project taking multiple years and sometimes multiple decades.
- ☐ **03** Large projects fail because they are so expensive as well as intrusive, overwhelming and disruptive to their environment that they often lead to new political problems. Even though a project may help the majority of people, there will always be people who're harmed by the project. These people legitimately complain, and they raise the question asking if it would be wiser to spend the money on another project.
- ☐ **04** The local people and governments will often lose their patience because grand projects taking decades lack the ability to show results and improvements during the process.
- ☐ **05** The world changes, and so the solution you thought of at the beginning may not be the right one after the first ten years of work. In fact, the problem may have changed so that you're now solving the wrong problem. Furthermore, technology changes and the local people's interest in applying changes will often change over a ten-year period.

Start Learning Today:
interaction-design.org

Creative Commons BY-SA license: You are free to edit and redistribute this template, even for commercial use, **as long as you give credit to the Interaction Design Foundation**. Also, if you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.

INTERACTION DESIGN FOUNDATION
Est. 2002

Do You Want to Learn More?

Learn how to use this template to your best advantage in our online course **Design for the 21st Century with Don Norman**. Sign up for it today and learn practical applications to improve your own work, if you haven't already started the course.

Design for the 21st Century with Don Norman

Beginner Course

In this course taught by Don Norman, co-founder and Principal Emeritus of Nielsen Norman Group, you'll learn **how designers can improve the world**, how you can apply human-centered design to **solve complex global challenges**, and what 21st century skills you'll need to **make a difference in the world**. Each lesson will build upon one another to increase your knowledge of human-centered design and provide you with practical skills to make a difference in the world. You'll also learn:

- How designers can improve the world
- How to use human-centered design to solve global challenges
- Why we evolved from User-Centered design to People-Centered design
- The difference between Wicked Problems and Complex Socio-technical Systems
- How to focus on people when you solve complex global challenges
- What feedback loops are and why they are so important
- How to tackle major problems by muddling through and taking small steps to generate successful innovative results

If you understand **why human-centered design is so important**, learn from its history and integrate its principles into your work, you will produce positive results that enhance peoples' lives. It can be done. So let's begin.

Learn more about how you can apply human-centered design to solve complex global challenges in the course **Design for the 21st Century with Don Norman**.

Learn more about this course >

Start Learning Today:
interaction-design.org

Creative Commons BY-SA license: You are free to edit and redistribute this template, even for commercial use, **as long as you give credit to the Interaction Design Foundation**. Also, if you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.

Interaction Design Foundation
Est. 2002

How to Advance Your Career With Our Online Courses

Take Online Courses by Industry Experts.

Lessons are self-paced so you'll never be late for class or miss a deadline.

Get a Course Certificate.

Your answers are graded by experts, not machines. Get an industry-recognized Course Certificate to prove your skills.

Advance Your Career.

Use your new skills in your existing job or to get a new job in UX design. Get help from our community.

See all our courses >

About the Interaction Design Foundation

With over 66,000 alumni, the Interaction Design Foundation is the **biggest design school globally**. Industry leaders such as IBM and Adobe train their teams with our courses, and universities such as MIT and the University of Cambridge include our courses in their curricula. Our online courses are taught by industry experts and cover the entire spectrum of UX design from beginner to advanced. We give you industry-recognized course certificates to advance your career. Since 2002, we've put together the world's biggest and most **authoritative library** of open-source UX Design literature created by such noted authors as Don Norman and Clayton Christensen.

INTERACTION DESIGN FOUNDATION

Start Learning Today:
interaction-design.org

Creative Commons BY-SA license: You are free to edit and redistribute this template, even for commercial use, **as long as you give credit to the Interaction Design Foundation**. Also, if you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.